
Teaching Guidelines for
Web-based Java Programming
PG-DAC August 2025

Duration: 112 hours (58 theory hours + 54 lab hours)

Objective: To learn advanced concepts in java programming and perform web Programming using Java.

Prerequisites: Knowledge of core Java programming

Evaluation: 100 marks

Weightage: CCEE – 40%, Lab exam – 40%, Internals – 20%

Text Book:

- Core and Advanced Java Black Book / Dreamtech Press

References:

- Servlet and JSP: A Tutorial by Budi Kurniawan / Brainy Software
- Spring in Action by Craig Walls / Manning Publications
- Advanced Java programming by Uttam K Roy / Oxford University press
- Sun Certified Enterprise Architect for Java EE Study Guide by Mark Cade & Humphrey Sheil / Pearson
- Professional Java EE Design Patterns by Murat Yener, Alex Theedom & Reza Rahman / Wrox

(Note: Each Session is of 2 hours theory and 2 hours lab unless mentioned otherwise)

Sessions 1 & 2:

Lecture:

JDBC & Transaction Management

- Introduction to JDBC API
- JDBC Architecture
- JDBC Drivers
- JDBC Classes & Interfaces: Driver, Connection, Statement, PreparedStatement, ResultSet and their relationship to provider implementations
- Stored procedures and functions Invocation
- SQL Injection overview and prevention
- Design Pattern: Data Access Object Pattern

Lab:

- Add Database CRUD operations to above MVC web application using JDBC Classes

Session 3:

Lecture:

J2EE Overview

- J2EE Container
- Packaging Web applications
- J2EE compliant web application

- Deployment tools.
- Web application life cycle
- Deploying web applications.
- Web Services Support

No Lab

Sessions 4, 5, 6 & 7:

Lecture:

- **Servlets: Dynamic Content Generation**
- **Advantages of Servlets** over CGI
- **Servlet Life cycle**
- **Servlet API & Deployment**
- **Servlet Annotations**
- **The Servlet interface**
- **The HttpServlet, HttpServletRequest, HttpServletResponse**
- **Exception Handling**
- **Servlet, DAO, POJO DB Layers**
- **Session**
- **Session Management**
- **Session Tracking with**
 - **Cookies**
 - **HttpSession**
- **Request Dispatcher**
- **Page Navigation**
- **Complete Case study Servlet Based**

Lab:

- Installing a servlet container (Tomcat)
- Adding Server to IDE
- Develop a structured dynamic web application(e.g. Library Management System) using servlets, deploy it in Tomcat
- Use HTTP Session in the Air Ticket Reservation System

Reading: Know more about the HTTP protocol at www.w3c.org

Tutorial: Compare which way of session tracking is better Cookies or HttpSession.

Sessions 8 & 9:

Lecture

- **JSP: Separating UI from Content generation code**
- **MVC architecture**
- **Design Pattern: MVC Pattern**
- **Life cycle of a JSP page**
- **Directives, Implicit and Explicit Objects, Scriptlets, Expressions, Expression Language**
- **Scope**
- **JSP Error Page handling**
- **JSTL**

Lab:

- Separate UI code from the controller code in your Library Management System by incorporating JSP and Servlets.
- Complete the implementation of Air Ticket Reservation System.
- Implement MVC based web application using Servlet, JSP

Sessions 10, 11, 12 & 13:**Lecture:**

- **Hibernate Framework**
 - Introduction to Hibernate Framework
 - Architecture
- **Hibernate in IDE**
 - Creating web application using Hibernate API
 - Lifecycle of Hibernate Entities
- **HB with annotation example**
- **Hibernate Mappings and Relationships**
- **Collection and Component Mapping**
- **HQL, Named Queries, Criteria Queries**

Lab:

- Demonstrate Hibernate as standalone library in Java application
- Develop a web application (Online Bookshop) using Hibernate Persistence

Reading: Study Hibernate architecture from www.hibernate.org/docs

Sessions 14, 15, 16 & 17:**Lecture:**

- Introduction to Spring Framework
- Overview of Spring Architecture
- Spring MVC architecture
- Spring Modules Overview
- Understanding Spring 4 annotations (Basic Introduction)
- IoC (Inversion of Control)
- IOC container
- Dependency Injection
- Spring Beans
- Autowiring Beans
- Bean Scopes
- Spring MVC
- Model, Model & View, HandlerMapping, ViewResolver
- Design Pattern: Front Controller Pattern
- Spring MVC Web application with JSP views (without Spring Boot)
- Using Thymeleaf as alternate View Technology (only introduction)
- Spring Validations
- Spring i18n, Localization, Properties
- File Upload example

Lab:

- Design and deploy Library Management System using Spring Web

Sessions 18, 19 & 20:

Lecture:

- Spring Boot essentials
- Need of Spring boot
- Spring Boot Overview
- Basic Introduction of MAVEN
- Building Spring Web application with Boot
- Spring Boot in detail (Use Spring Boot for all demo & assignments here onwards)
- Running a web application using Spring Boot with CRUD (with Static Data not DB)

Lab:

- Create Hello World Spring Boot Web application
- Check Libraries imported by Spring Boot
- Create Spring Boot CRUD application with Thymeleaf as View technology

Sessions 21 & 22:

Lecture:

Spring Data Module

- Spring Data JPA (Repository support for JPA)
- CrudRepository & JpaRepository
- Query methods
- Using custom query (@Query)

Lab:

- Add CRUD operations with Spring JPA etc. to earlier Spring Web application.

Session 23:

Lecture:

Spring AOP

- AOP Overview
- Spring AOP
- AOP Terminology and annotations: Advice, Join Points, Pointcuts, Aspects

Lab

- Modify earlier Spring MVC application to Log all the requests using AOP

Sessions 24 & 25:

Lecture:

Building REST services with Spring

- Introduction to web services
- SOAP Vs RESTful web services
- RESTful web service introduction
- Create RESTful web service in java using Spring Boot
- RESTful web service JSON example
- RESTful web service CRUD example
- Using POSTMAN client to invoke REST API's
- REST service invocation using REST Template

Lab:

- Create REST API for Employee Management using Spring Boot
- Invoke it from POSTMAN app
- Invoke it from another Spring Boot Web application using REST Template

Session 26 & 27:

Lecture:

Testing in Spring

- Testing in Spring
- Unit Testing of Spring MVC Controllers
- Unit Testing of Spring Service Layer
- Integration Testing of Spring MVC Applications: REST API
- Unit Testing Spring MVC Controllers with REST

Securing Web Application with Spring Security

- What is Spring Security
- Spring Security with Spring Boot
- Basic Authentication
- Authentication with User credentials from Database and Authorization
- JWT Authorization

Lab:

- Design & Test Spring Application
- Secure the Spring Web application created in earlier exercise.

Sessions 28 & 29:

Lecture:

Microservices

- Introduction to Microservices
- Microservices Architecture
- Fragmentation of business requirement
- Deployment pattern
- API gateway
- Service Discovery
- Database Management for Microservices

Lab (2 hours):

Demonstration of Spring Boot Microservices